Optimal Roster Constitution in Basket-Ball.

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Date TBD

Abstract

To Do

1 Introduction

Sports analytics has been a burgeoning field for decades. In team sports, data were primarily descriptive and high-level, focusing on direct measurements necessary to determine the outcome of a contest (number of points scored, win-loss records, etc.). The mid-20th century, through the professionalization of sports, saw the emergence of individual performance statistics, such as shots, tackles, and passes. These data, commonly referred to as **box-score** statistics, brought a player-centric dimension to performance analysis. Indeed, they introduced the possibility of attributing collective performance to individual players. Later development, with the increasing use of tracking systems in the 2010s, allowed to considerably enrich these data, adding more granularity. In the National Basketball Association (NBA), a real-time tracking technology, SportVU, was introduced into every arena in 2014. It allows differentiating shots no longer by their sole value (2 or 3 points) but by their location (elbow, post, etc.), or assess if a rebound is or is not contested.

While these statistics paved the way for aggregated indicators used to deepen the understanding of analyzed sports, with Sabermetrics in baseball (James, 1984) and Oliver's Four Factors in basketball (Oliver, 2004) as pioneering efforts, they remain widely used to assess individual performance. In the NBA, arguably the most competitive basketball league in the world, individual awards such as Most Valuable Player (MVP), Defensive Player of the Year (DPoY), selections to the NBA All-NBA Teams, and participation in the All-Star Game are to a great extent determined by players' statistical output. In short, box-score statistics are routinely used to identify and evaluate key players.

However, in team sports, while individual abilities and athletic level are crucial factors of performance, it is ultimately teams that win games and championships. Bridging the gap between individual performance analysis and team performance therefore represents a central challenge for sports analytics. More specifically, a fundamental question arises: how does the value of a team compare to the sum of the values of its individual constituents? And, conversely, how can one infer, from individual statistics alone, the value of a potential team? A seminal illustration of the use of box-score statistics to assemble a competitive team was presented in the book *Moneyball: The Art of Winning an Unfair Game*, later adapted into a major motion picture. This story describes how a small-budget baseball team managed to leverage Sabermetrics to build a competitive roster despite significant financial constraints. While *Moneyball* recounts a successful application of box-score-derived statistics, granting such data an overestimated relevance may yield opposite results. Since the 2010s, the NBA has witnessed several attempts at assembling so-called "Super Teams", few of which achieved sustained success. While no strict consensus on the term "Super Team" has been

established in the academic literature, NBA.com writer Steve Aschburner proposed the following defining characteristics[1]:

i It boasts two, three or more stars on its roster, any one of whom would be able to anchor a franchise on his own.

ii Trades and/or free agency had to have been used to unite those stars, either in their prime or at least after obvious, recognized success (All-Star appearances, all-NBA selections, major awards won).

iii The team needs to be a reasonable contender, not merely a collection of big names past their prime and far back of the reigning conference or league champs.

iv Bonus points if media outlets have carried one or more stories about the alleged Super Team wondering (a) if one basketball is enough, or (b) how the stars' strong personalities might gel or repel.

The recurring underperformance of such assemblies of individually dominant players relative to expectations highlights the limits of individual statistics as a sole proxy for team success, and motivates the development of more holistic analytical frameworks.

This work proposes a data-driven method, leveraging box-score statistics, to address the task of roster construction in basketball. Our methodology unfolds in two complementary steps:

1. We apply **unsupervised learning** to players' individual statistics in order to identify archetypes of players, thereby creating an intermediate level of analysis that transcends the athlete level, at which each individual is unique by definition. To this end, we make use of a dataset referenced in Muniz and Flamand (2022)[2].

2. We subsequently leverage these archetypes to **characterize each team according to the lineups it deploys**, and relate this characterization to team success. Lineups were extracted from play-by-play data, publicly available on Kaggle[3].

The remainder of this paper is organized as follows: Section 2 reviews related work on unsupervised learning applications in sports analytics and roster construction, Section 3 describes our methodology, Section 4 presents our results, which are discussed in Section 5, and Section 6 concludes the study.

2 Related Work

2.1 Unsupervised Learning in Sports

Machine Learning has been successfully applied for over a decade in sports analytics (Memmert, 2025). In this section, we are focusing on application of **unsupervised learning**. These approaches are primarily utilized to address the inherent lack of ground-truth labels in professional tracking data, focusing on player archetype discovery, tactical interaction modeling, and game-phase segmentation.

[1] https://www.nba.com/news/nba-super-teams-and-history, consulted on February 25th, 2026

[2] https://github.com/meganlynn12/jsa-cdkm?tab=readme-ov-file, consulted on February 25th, 2026

[3] https://www.kaggle.com/datasets/schmadam97/nba-playbyplay-data-20182019, consulted on February 25th, 2026

Discovery of Player Roles and Archetypes Unsupervised learning provides a robust mechanism for redefining player positions based on empirical performance rather than traditional categorical labels. Muniz and Flamand (2022) introduced a novel two-step methodology utilizing k-Means clustering to initialize similarity weights within a weighted network, followed by the Louvain method for community detection. Applied to NBA basketball tracking data (2014–2020), the study identified eight distinct archetypes, such as "Superstar Guards" and "Role Playing Bigs", which explained significantly more variation in team success (Win Shares) than the traditional five-position model. While this study relates closely to our archetypes optimization objective, if differs as clusters are different across seasons, when we need comparable clusters to consistently characterize teams' lineups. In soccer, Bialkowski et al. (2014) utilized minimum entropy data partitioning, an Expectation-Maximization (EM) approach to detect team formation in presence of player permutations. By learning roles directly from spatiotemporal data, they found that a dynamic role-based representation allows for accurate formation detection and provides necessary context for analyzing individual player behavior that is lost in static identity models.

Tactical Interactions Miller and Bornn (2017) implemented a hierarchical model involving probabilistic clustering via Bezier curves and Latent Dirichlet Allocation (LDA) to discover an offensive "vocabulary" in the NBA. This unsupervised approach successfully grouped similar possessions, such as "weave" plays or specific tactical sequences leading to corner three-pointers, enabling efficient search and exploration of vast tracking databases. Knauf et al. (2015) developed a novel class of Spatio-Temporal Convolution Kernels (STCKs) applied through kernelized k-medoids and spectral clustering. Utilizing Bundesliga soccer data, they demonstrated that these kernels effectively identify recurring movement patterns such as "game initiations" and "scoring opportunities". Their findings highlight that STCKs are capable of distinguishing between multifaceted team philosophies, such as the rehearsed short moves of one team versus the linear, long-ball strategies of another, more effectively than baseline Hausdorff distance or bag-of-words methods. In handball, Schrapf et al. (2017) employed self-organizing maps (specifically the DyCoN neural network) to cluster offensive and defensive constellations. They found that specific defensive patterns were significantly more successful (measured via odds ratios) against certain offensive behaviors, providing an objective framework for preparation against opponents. Leser et al. (2019) applied spectral clustering following Laplacian Score-based feature reduction to analyze youth soccer 3-vs-2 games. This unsupervised mining identified "hidden" tactical strategies, revealing that offensive efficiency is maximized when attacks are initiated via the wings and concluded with a backward "through-ball" pass against the playing direction. Fewell et al. (2012) treated NBA teams as dynamic networks, using clustering coefficients and network entropy to quantify offensive strategies. They found that team-level success was positively correlated with higher network entropy (unpredictability) and lower degree centrality, suggesting that elite teams tend to distribute decision-making across multiple players rather than relying on a single dominant ball-handler.

Game Phase Segmentation Perše et al. (2009) utilized Gaussian Mixture Models (GMM) and the EM algorithm to segment basketball trajectories into game phases (offense, defense, and timeouts) with approximately 90% accuracy. Furthermore, they used k-means clustering to partition the court into semantic regions, enabling the automatic recognition of organized play types using Levenstein distance between observed actions and database templates. Montoliu et al. (2015) proposed a Bag-of-Words (BoW) strategy to recognize activities in soccer without requiring individual player tracking. By extracting "motion words" from optical data, they achieved high classification accuracy (up to 92.8%) for team activities such as Ball Possession, Quick Attack, and Set Pieces.

2.2 Roster Optimization

The application of machine learning and mathematical optimization has become a cornerstone of modern sports analytics, shifting the focus from individual talent to the synergistic effects of player combinations. In soccer, Abidin (2021) implemented several to evaluate players based on training data and coach assessments. The study demonstrated that Random Forest models could select optimal player combinations with 93.9% reliability, producing lineup suggestions that were 97.2% similar to a professional coach's ideal team. In Basketball, Ke et al. (2024)[4] developed a unified framework for the NBA and WNBA (United States' professional women basketball league) that combined unsupervised clustering (K-Means and Hierarchical Agglomerative Clustering) for player categorization with supervised neural networks to predict the winning probability of specific rosters based on playing time proportions. Similarly, Penner (2025) moved toward positionless basketball by using K-means clustering to identify nine distinct player archetypes across 110 global leagues, subsequently utilizing multiple linear regression to rank athletes by "points per minute" to optimize championship roster construction. In the specialized context of wheelchair basketball, Calvo et al. (2025) combined Bayesian longitudinal modeling with stochastic integer linear programming to identify optimal lineups while strictly adhering to the Player Classification Points System (PCPS) constraints. This methodology allows coaches to incorporate uncertainty into the selection process, effectively estimating the probability that a five-player configuration will yield the highest team performance. Beyond traditional field sports, Sapienza et al. (2019) explored team composition in the Multiplayer Online Battle Arena games Dota 2, employing deep neural autoencoders to recommend teammates who maximize an individual's "TrueSkill" score gain. Their model achieved a 19.5% improvement in recommendation accuracy over traditional graph factorization methods. Such approaches are supported by theoretical frameworks like those proposed in Liemhetcharat and Veloso (2012), which utilize synergy graphs and simulated annealing to model how task-based distances between heterogeneous agents affect overall team capability.

3 Methodology

3.1 Data

We relied on three data sources in this paper: (i) a tracking-improved box-score dataset for identifying players' archetypes (see paragraph 3.1.1), (ii) play-by-play data to retrieve lineups used by teams and their relative +/- statistics (see paragraph 3.1.2) and (iii) team performance metrics to assess success of teams (see paragraph 3.1.3). We only retain in our analysis complete seasons where these three sources are available, and considered only regular season games. Playoff games were discarded, as we believe they could bring a distortion to the analysis. Indeed, only the best performing teams of a season qualify to the playoffs, facing stronger adversity, which could deteriorate their performance. Moreover, this choice allows to have exactly the same number of games for all NBA teams, and therefore not over represent good performing teams. Our dataset covers 5 NBA seasons, from 2015/16 to 2019/20, representing a total of 5,980 games (1,230 per season from 2015/16 to 2018/19 and 1,060 for the 2019/20 season, due to the modification of the season's schedule incurred by the Covid-19 pandemic). An overview of the datasets can be found in Table 1.

3.1.1 Tracking-improved box-score data

The dataset presented in this paragraph has been referenced in Muniz and Flamand (2022) and is publicly available on their github repository[5]. It consists in data obtained from the website

[4] On le cite alors que c'est le papier avec erreur, comment gérer ?

[5] Consulted on February 25th, 2026

Table 1: Datasets overview

Season	Nb games	Nb players*	Nb Play-by-play lines	Nb lineups
2015-16	1,230	472	601,557	12,759
2016-17	1,230	484	596,645	12,960
2017-18	1,230	532	590,868	13,869
2018-19	1,230	508	614,516	14,761
2019-20	1,060	523	539,265	13,811
Total	**5,980**	**2,519**	**2,942,851**	**68,160**

* Notes: (i) the total number of player does not correspond to unique players as players can play several seasons, (ii) we discarded players that were missing in one or more datasets. None of them play more than 10 minutes per game on average.

nba.com, initially split in 21 raw datasets and reorganized by the authors of Muniz and Flamand (2022) in 6 datasets as described in Table 2. Please note that contrarily to Muniz and Flamand, we did not consider the "Clutch" dataset as it contains data from particular part of games (less than 5 points difference in the last 5 minutes), and therefore overlaps with other datasets. These data contains traditional basketball box-score statistics (e.g. shot attempted, shot made, rebound, assists, etc.) augmented from Sports VU tracking system (introduced in NBA arenas in 2014). Tracking improved box-score statistics as it (i) brought higher granularity to existing indicators (for instance, categorizing shots by areas on the pitch rather than sole value, identifying if a rebound or a shot is contested or not, etc.) and (ii) created new indicators (by recording for instance the number of boxouts[6]).

Table 2: Tracking-improved dataset description

Dataset	Description	Raw Source(s)
Scoring	2/3 FG(M/A/%)	Shooting
	(Drive/Catch Shoot/Pull Up/Paint/Post/Elbow) Touch PTS and %	Tracking Efficiency
	PTS per (elbow/post/paint) touch	Tracking Possessions
	PTS (off TOV/2nd chance/fastbreak/in the paint)	General Misc
	USG%, Off. Rating	General Traditional
	DD2, TD3	General Traditional
Passing/ Playmaking	AST %, AST:TO, AST Ratio	General Advanced
	Time of possession, touches (all & front-court), avg. sec & dribble per touch	Tracking Possessions
	Drive passes, AST, TOV	Tracking Drives
	AST, FT AST, Secondary AST, Potential AST, AST:Pass	Tracking Passing
	Pass/AST/TOV per (elbow, post, paint) touch	Tracking Elbow, Post, Paint Touch
	Screen AST	Hustle
Rebounding	OREB/DREB (Contest/Uncontest/Chance/Distance/Defer)	Tracking Rebounding
	Off/Def Box-outs, Box-out REBs	Hustle
Defense	STL, BLK, Def Rim FGM/A	Tracking Defense
	Def FGM/A, % difference (FG%)	Tracking Defensive Impact
	Opp 2nd chance/fastbreak/in-paint points	Tracking Defense
	Deflections, Charges Drawn, Contested 2/3 PT shots	Hustle
Hustle/ Miscellaneous	Loose balls recovered	Hustle
	Personal fouls drawn	General Traditional
	Drive personal fouls	Tracking Drives
	Distance traveled, average speed	Tracking Speed Distance

Adapted from Muniz and Flamand (2022). For an extensive description of the statistics, please refer to https://www.nba.com/stats/help/glossary

In these datasets, every entry corresponds to a player's average statistics over a season. We concatenated seasons and datasets and removed of duplicated columns (minutes and number of

[6]A boxout is a situation where a player manages to position himself between the basket and his opponent in a shooting situation, to achieve higher probability of securing the rebound.

games played are recorded in each dataset). In total, the concatenated dataset contains 2,519 entries (player's season) distributed into 30 NBA teams, and has 129 dimensions (performance indicators). We manually excluded features being collinear by design (e.g. TS_{PCT}, the True Shooting Percentage, is obtained from other shooting statistics, AST and TOV, the total number of assists and turnovers is the sum of the number of assists and turnovers by locations, etc.), which reduced the number of features to 116. We subsequently scaled our dataset using a TBD scaler and further reduced the dimensionality of our dataset by applying a Principal Component Analysis (PCA) and retaining 80% of explained variance.

3.1.2 Play-by-play data

The dataset presented in this section is publicly available Kaggle[7]. Play-by-play data record timestamp-indexed of event in a game. Recorded event encompasses traditional box-scores events (e.g. shot attempts, rebounds, assists, etc.) and the identity of the players credited for the corresponding events, as well as game-related information (start of period, end of period, player substitutions, etc.). From the identity of players credited for event and the changes log, we were able to identify for all timestamps the composition of both team participating in a game. Plus de détails sur le process ? A composition is a combination of 5 players from the pool of available players. In the NBA, teams are allowed to register 12 players for a given game, corresponding to $\binom{12}{5} = 792$ possible combinations. In practice, In total, we identified between 12,759 and 14.761 unique lineups per season. Teams aligned between 187 and 794 different lineups over the course of a season ($\mu = 454.4, \sigma = 114.7$).

3.1.3 Team performance data

We collected from nba.com two indicators of teams' success:

i Teams' Win Rate (number of games won / number of games played), ranging from 0.122 to 0.890 ($\mu = 0.500, \sigma = 0.152$).

ii A binary indicator worth 1 if the considered team reached the conference semi-finals in the playoffs and 0 otherwise, as suggested in Ke et al. (2024). The NBA is split into two conferences (East and West), both having their own playoff competition. The NBA champion is the winner of the best-of-seven series between East and West champions. Reaching the conference semi-finals is therefore equivalent to being within the last 8 contenders for the NBA title, which corresponds to the top 26.7%.

An illustration of the distribution of success can be found in Figure 1. Top performing teams appear to be the Golden State Warriors 2015, 2016, 2017 and the San Antonio Spurs 2016, achieving win rates above 80% and reaching Conference semi-finals, while Minnesota Timberwolves 2015 and Philadelphia 76ers 2016 emerge as low performers with win rate below 20%. Atlanta Hawks 2017 and 2019, and Memphis Grizzlies 2017 and 2018 aligned over 700 different lineups during the season, and Minnesota Timberwolves 2017, Orlando Magic 2018 and Detroit Pistons 2015 less than 240. None of these teams reached the Conference semi-finals.

3.2 Clustering of NBA players

From the tracking-improved boxscore dataset (see 3.1.1), we aim at identifying players' archetypes. To do so, we need to create a partition of our dataset. We benchmarked various approaches, including centroid-based, hierachical agglomerative, density-based and gaussian models, with a hyperparameters search.

[7] Consulted on February 25th, 2026

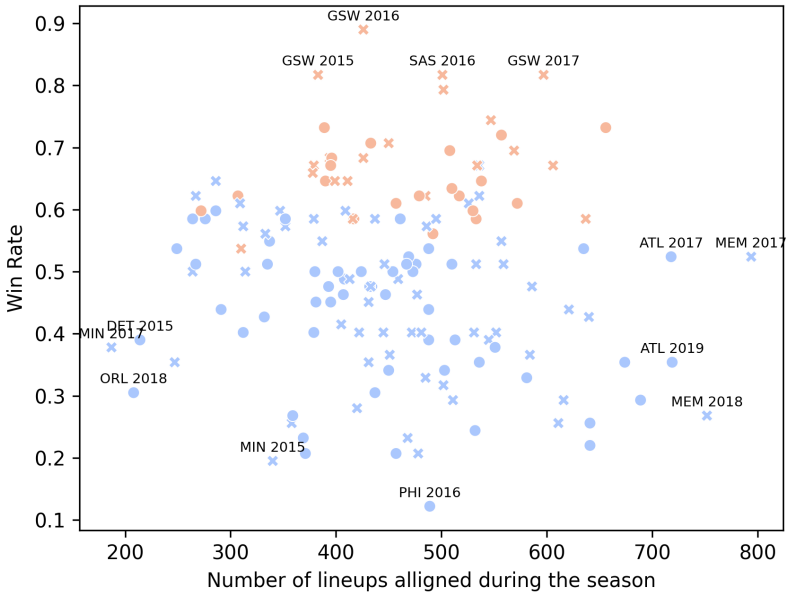

Figure 1: Distribution of teams according to their number of lineups aligned during the season (x-axis) and win rate (y-axis). Red dots indicate teams reaching the Conference semi-finals, and blue dots the opposite, Crosses correspond to Western Conference teams and dots to Eastern Conference teams.

Among partition-based approaches, the centroid-based k-means algorithm partitions observations into k groups by minimizing within-cluster dispersion, typically defined as the sum of squared Euclidean distances between observations and their corresponding cluster centroids. Hierarchical clustering methods provide an alternative framework by constructing nested partitions without directly optimizing a global objective. In particular, agglomerative hierarchical clustering proceeds iteratively by merging pairs of clusters according to a linkage criterion (single, complete, average, or Ward's minimum-variance linkage). The resulting dendrogram encodes a multi-scale representation of the data, allowing cluster structure to be examined at different levels of granularity. Density-based approaches define clusters as connected regions of high sample density. HDBSCAN extends the DBSCAN framework by building a hierarchy based on mutual reachability distances and extracting clusters according to their stability across density levels. This formulation enables the identification of clusters with varying densities while treating low-density regions as noise, which is particularly relevant in high-dimensional and irregularly distributed datasets. However, in our case, we allocated "noisy" players to the nearest cluster (based on distance to cluster's centroid), as (i) we see no reason to discard outliers and (ii) there is a significant risk high performing (and therefore highly represented) players would be considered as outliers. Finally, model-based clustering relies on probabilistic generative assumptions. Gaussian Mixture Models (GMMs) assume that observations arise from a finite mixture of Gaussian distributions, each parameterized by a mean vector and covariance matrix. Model parameters are typically estimated via the EM algorithm, yielding soft probabilistic assignments of observations to latent components.

Details about the methods, hyperparameters searched and hyperparameters values can be found in Table 3.

Comparison metric To determine which clustering algorithm yields the best results on our dataset, we need to compare them. Contrary to supervised learning, where a form of consensus emerges around some key metrics (e.g. Accuracy, F1 score and ROC-AUC for classification

Table 3: Algorithms and Hyperparameters tested

Algorithm	Hyperparameters	Range
K-Means	Number of clusters	{2, ..., 12}
	Type	Basis, Spherical
Agglomerative	Number of clusters	{2, ..., 12}
	Linkage function	Ward, Average, Complete
HDBSCAN	Minimum Cluster Size	{4, 6, 8, 10, 12, 16, 20, 30, 40, 50}
	Minimum Samples	{3, 5, 7, 10, 15}
	Cluster Selection Epsilon	{0, 0.2, 0.5, 1, 2}
	Maximum Cluster Size	{0, 400, 800}
GMMs	Covariance Type	Spherical, Diagonal, Full, Tied
	Number of components	{2, ..., 12}

tasks, RMSE and its variants for regression tasks), Clusteri Validity Indices (CVIs) are still subject to discussion within the academic community. Arbelaitz et al. (2013) compared 30 CVIs and concluded three groups of indices emerge, and the first of them contains indices that "behave better", with "the differences being statistically significant". Well know metrics such as Silhouette Score, Davies-Bouldin, Calinski-Harabasz, etc. are listed as part of this well performing group. On the other hand, Gagolewski et al. (2021) stated it is better "not to treat them as objective functions for identifying good partitions", and specified this is particularly the case for Silhouette and Davies-Boulin. We therefore chose to rely on the Calinski-Harabasz index to compare different methods. This metrics is defined as follows:

$$CH = \frac{n-k}{k-1} \times \frac{\sum_1^n \|\mu - \mu_{C(i)}\|^2}{\sum_1^n \|x_i - \mu_{C(i)}\|^2} \tag{1}$$

where n is the population size, k the number of clusters, $C(i)$ is the clusters in which point $x_i, i \in \{1, ..., n\}$ is allocated, μ and $\mu_{C(i)}$ are the centroids of the population and of cluster $C(i)$ respectively, and $\|a - b\|$ represents the Euclidian distance between points a and b.

Revoir Moreover, we will (i) use a visual "elbow method" to select our optimal hyperparameters, favoring a higher number of clusters in case of close performances and (ii) perform a manual "coherence check" to verify clusters relevance

3.3 Lineups analysis

To do

4 Results

4.1 Clustering

The Calinski-Harabasz indices of the compared method are presented in Figure 2. Please note that the number of clusters if for some algorithms (e.g. K-Means, GMM) an input hyperparameter, but for some others (e.g. Partition - HDBSCAN), it is a consequence of other hyperparameters. A clear hierarchy emerges between algorithm : K-Means systematically outperforms other algorithms, while Agglomerative with Complete and Average linkage emerge as worst performers. Agglomerative with Ward linkage and GMM complete the top 3, the former performing better with few clusters and the latter with more clusters. Partition HDBSCAN and Spherical K-Means relative performance deteriorates with higher number of clusters, the former achieving a local maxima with 5 clusters, with a score comparable to GMM. Interestingly, performances of all methods

decrease with the number of clusters, consistently for most algorithms, and with the presence of some local maxima for others (Partition HDBSCAN, Agglomearative with Average and Complete linkage).

To do: visual inspection: choosing k from elbow

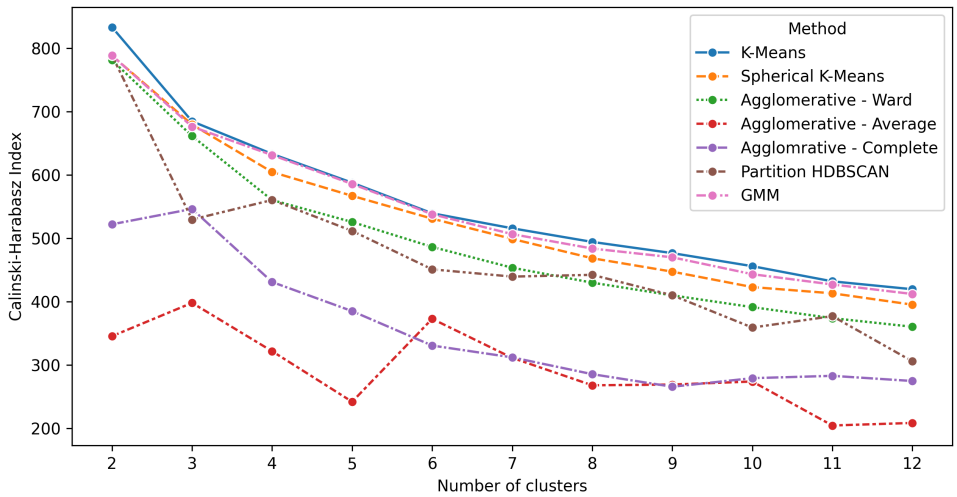

Figure 2: Comparison of clustering algorithm.

Figure 3 presents a visual inspection of the selected partition.

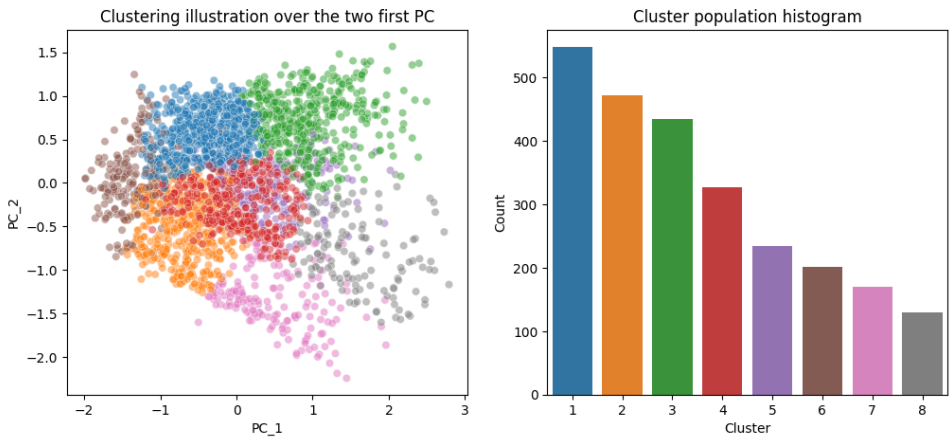

Figure 3: Visualization of selected partition.

4.2 Roster Constitution

To do

5 Discussion

To do

6 Conclusion

To do

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